# Dispersal and metapopulations

## Learning objectives:

1. Understand how populations persist in disturbed landscapes
2. Know what local dispersal is and how to model it
3. Build a metapopulation model that includes dispersal and disturbance

*Environmental heterogeneity, including spatial, can affect population growth rate*

*Connect multiple populations in a heterogeneous environment*

## Patch dynamics

### Source-sink dynamics:

- Caused by environmental heterogeneity
- Good patches/bad patches -> source-sink dynamics

*Draw lattice with patches*

- Source patches: F > m *excess of seeds*
- Sink patches: F < m

*Circle plants in patches*

- metapopulations: population of populations

### Disturbance:

- homogeneous environment, regular disturbances
- disturbances can affect some patches but not others
  - example: falling tree disturbed understory, patchy fire, storms
- seed banks to avoid periods of disturbance

*question: how to avoid patchy/spatial disturbances that could wipe out a seed bank? what's required so they don't go extinct?*

## Dispersal

- local vs. global dispersal
- closer to parent = more seeds
- dispersal kernels - 3D distributions

*Draw circle around individual plant in lattice*

*Plot distance from parent with bell curve*

- types of dispersal kernels *draw each*
  - Gaussian
  - exponential
  - fat-tailed

- dispersal limitation - lower population growth in patchy/disturbed environments

### Competition-colonization tradeoff

- homogeneous environment, regular disturbances
- tradeoff between competitive ability and dispersal ability
- colonizer advantage after disturbance, competitor advantage over time
- intermediate disturbance = highest diversity

## Modeling

$$N_{t+1} = N_t(1-g)s + \frac{fgN_t}{1+\alpha g N_t}$$

Number of seeds produced:

- $p$ - focal patch
- $F_p$ - number of seeds produced in a given patch
- $N_p$ - population size for a given patch
- $f_p$ - fecundity within a patch, proportion of $f$

$$F_p = \frac{f_p g N_{p,t}}{1+\alpha g N_{p,t}}$$

- $P$ - number of patches
- $d$ - fraction of seeds that leave the patch

Seeds retained in patch:

$$F_p(1-d)$$

Seeds coming in from other patches:

- $j$ - all other patches aside from focal patch

$$\frac{d}{P-1}\sum_{j=1,j\neq p}^{P} F_j$$

$$N_{p,t+1} = N_{p,t}(1-g)s + F_p(1-d) + \frac{d}{P-1}\sum_{j=1,j\neq p}^{P} F_j$$

*this all assumes a homogeneous environment (patches are all equal), so we have to add disturbance*

Adding another element: disturbance

- sometimes patch $p$ gets wiped out so $N_p = 0$
- probability of disturbance, binomial distribution, 0 - no disturbance, 1- disturbed

patch abundances at time t: $N_1 = 20$, $N_2 = 13$, $N_3 = 8$

disturbance: 0, 1, 0

new patch abundances at time t: $N_1 = 20$, $N_2 = 0$, $N_3 = 8$